

Regularization for Deep Networks

Scaler School of Technology | Deep Learning I | Week 3 — Session 1/2 | 14 May 2026



What Is This Lecture About?

Deep networks have millions of parameters and can easily memorize training data instead of learning useful patterns. This session covers the essential toolkit of regularization techniques — the tricks that make your models **generalize** to new, unseen data.

1 Overfitting & Weight Decay

Why deep nets overfit so easily (they can even memorize random labels!), and how penalizing large weights with L2 regularization keeps the model simple. *Think of it as giving the network a “budget” for its weights.*

2 Dropout

The famous trick of randomly “switching off” neurons during training. Forces the network to develop redundant, robust features. *Like a team project where random members are absent each meeting — everyone has to learn everything.*

3 Early Stopping & Data Augmentation

Know when to stop training (before overfitting gets worse) and how to create “free” training data through label-preserving transformations like flips, rotations, and crops.

What to Expect in the Session

- Live demo: overfitting a network on a toy dataset
- Visualizing decision boundaries with/without regularization
- ★ Seeing a network memorize **random labels** perfectly
- Hands-on: comparing dropout rates in a notebook
- 2 quizzes + 1 group discussion
- A decision framework: “which regularizer when?”

How to Prepare (Optional but Helpful)

- ✓ **Recall** what overfitting looked like in your ML course (bias-variance tradeoff)
- ✓ **Recall** Ridge Regression — L2 regularization is the same idea
- ✓ **Think about:** “If a model has 11M parameters and only 50K training samples, what could go wrong?”
- ✓ **Skim** PyTorch docs for `nn.Dropout` if curious about the API

Duration: ~2 hours **Prerequisites:** Backpropagation, loss functions, SGD/Adam **Instructor:** Priyansh Saxena